

Section: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3102**

Title: **STRENGTH OF METALS**

Revision: **1.1**

Date: **23/01/2019**

1 INTRODUCTION

1.1 METHOD-SUMMARY

This chapter sets Tresca and/or Von Mises stresses as the applicable point stress failure criteria for ultimate strength analyses of metallic ductile materials (note buckling/stability issues are not covered by this check).

1.2 SCOPE AND LIMITATIONS

- ✓ Metallic ductile materials
- ✓ Pure ultimate strength
- ✓ Bucking/stability not covered

2 GENERAL POINT STRESS FAILURE CRITERIA

Even fracture mechanics approaches may be useful in some detailed non-linear numerical FEM models, or in specific damage tolerance analyses, point stress failure criteria are preferred, for obvious reasons of simplicity, in most static strength checks.

- No permanent deformation (at LL) check to be done comparing the minimum between F_{ty} and F_{cy} (unless the sign of the applied stress is clear) with the Von Mises applied stress.

Note: Von Mises criterion is also known as the maximum distortion strain energy criterion. Von Mises stress can be also referred as the equivalent tension stress or the maximum octahedral shear stress.

- Ultimate strength in stable (non-buckled) parts shall be checked comparing F_{tu} with the Von Mises applied stress. Von Mises has been systematically applied and experience proved its

suitability (for practical reason it is preferred to Tresca / maximum shear failure criterion described below).

For plane stress, in terms of principal stresses (see cheatsheet below), these failure criteria can be expressed as:

$$\text{Tresca: } |\sigma_1| < F_{tu}, |\sigma_2| < F_{tu}, |\sigma_1 - \sigma_2| < F_{tu}$$

$$\text{Von Mises: } (\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2)^{0.5} < F_{tu}$$

Several references support approach above...

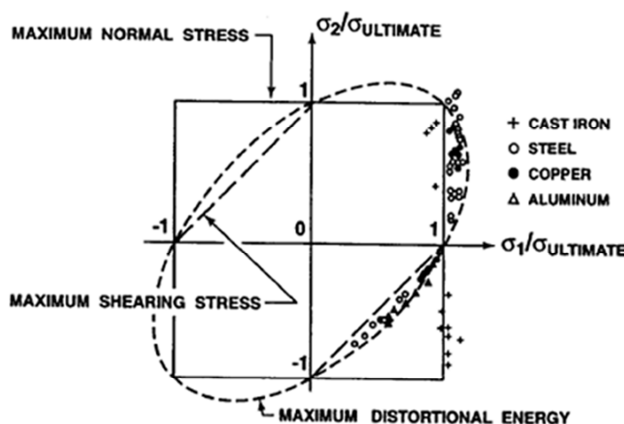
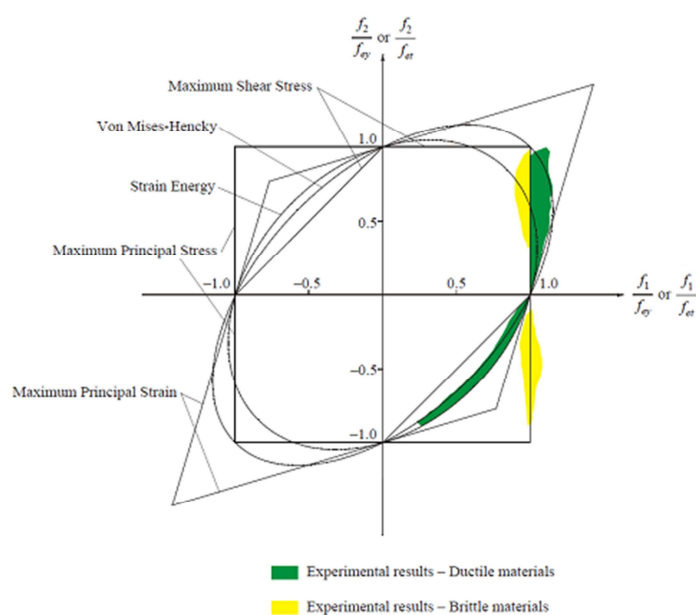


Figure 1. Jones', 2nd Edition, page 104 (Ref. 1)



Failure criteria above, and resulting static margins, are applicable to stable parts. Review chapters dedicated to buckling and stability in case the parts analyzed are sustaining relevant compression and/or shear loads.

3 CHEATSHEET: FORMULAS FOR RELEVANT STRESS INVARIANTS

3.1 Von Mises

The following table summarizes von Mises criterion for the different stress conditions.

Load scenario	Restrictions	Simplified von Mises equation
General	No restrictions	$\sigma_v = \sqrt{\frac{1}{2}[(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2 + 6(\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2)]}$
Principal stresses	No restrictions	$\sigma_v = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$
General plane stress	$\sigma_3 = 0$ $\sigma_{31} = \sigma_{23} = 0$	$\sigma_v = \sqrt{\sigma_{11}^2 - \sigma_{11}\sigma_{22} + \sigma_{22}^2 + 3\sigma_{12}^2}$
Principal plane stress	$\sigma_3 = 0$ $\sigma_{12} = \sigma_{31} = \sigma_{23} = 0$	$\sigma_v = \sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2}$
Pure shear	$\sigma_1 = \sigma_2 = \sigma_3 = 0$ $\sigma_{31} = \sigma_{23} = 0$	$\sigma_v = \sqrt{3} \sigma_{12} $
Uniaxial	$\sigma_2 = \sigma_3 = 0$ $\sigma_{12} = \sigma_{31} = \sigma_{23} = 0$	$\sigma_v = \sigma_1$

Table 1. Von Mises equivalent stress for several stress states

Notes:

- Subscripts 1,2,3 can be replaced with x,y,z, or other orthogonal coordinate system
- Shear stress is denoted here as σ_{ij} in practice it is also denoted as τ_{ij}
- Principal stresses are denoted as σ_i , or as $\sigma_{\max}$ and $\sigma_{\min}$

3.2 Tresca

The expression for the maximum shear stress is:

$$\tau_{\max} = (\sigma_{\max} - \sigma_{\min})/2$$

And the general form of Tresca Criterion (yield , failure):

$$\sigma_{\max} - \sigma_{\min} = F_y (, F_{tu})$$

For the common *plane stress* case the principal stresses are more simple to obtain, and the application of the Tresca criterion becomes straightforward. Let σ_1 , σ_2 and σ_3 be the principal stresses ($\sigma_3 = 0$). If σ_1 and σ_2 are of opposite sign, yield (, failure) occurs when:

$$\sigma_1 - \sigma_2 = F_y (, F_{tu})$$

If σ_1 and σ_2 carry the same sign, yield (, failure) occurs when:

$$|\sigma_1| = F_y (, F_{tu})$$

$$|\sigma_2| = F_y (, F_{tu})$$

3.3 Principal Stresses

The expressions to obtain the principal stresses for the two-dimensional cases of plane stress and plane strain are the following:

$$\sigma_1, \sigma_2 = (\sigma_x + \sigma_y) / 2 \pm \{ [(\sigma_x - \sigma_y) / 2]^2 + \tau_{xy}^2 \}^{0.5}$$

And the orientation of the principal planes using the formula:

$$\tan(2 \cdot \theta_{p1,2}) = 2 \cdot \tau_{xy} / (\sigma_x - \sigma_y)$$

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
e	%	Elongation, permanent deformation after tension failure
E	MPa	Young's modulus
F _{cy}	MPa	Yield compression stress
F _{tu}	MPa	Ultimate tensile stress
F _{ty}	MPa	Yield tensile stress
F _y	MPa	Yield stress (not specifying if in tension or compression)
Rb		Ratio between applied bending and bending allowable moment of a material full section
RF		Reserve Factor, i.e. ratio between allowable load and applied load
Rs		Ratio between applied shear and shear allowable load of a material full section
Rt		Ratio between applied tension and tensile allowable load of a material full section
σ	MPa	Axial stress
σ _v	MPa	Von Mises stress
τ	MPa	Shear stress
T _{max}	MPa	Tresca, or maximum shear, stress

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[Robert M. Jones]	2 nd	MECHANICS OF COMPOSITE MATERIALS
Ref. 2	ESDU Struct 02.00.00	1969	Criteria of yielding and failure
Ref. 3	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 4	HSB 51200-01	Is. E	Reserve factors from interaction curves for combined loading
Ref. 5	HSB 24000-01	Is. E	Allowable loads of bolts (general description)
Ref. 6	HSB 51100-01	Is. B	Equivalent stresses for different stress states
Ref. 7	[J. Chakrabarty]	3 rd Ed.	Theory of Plasticity

SOFTWARE

N/A

DISTRIBUTION LIST

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Table 1. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 01/08/2018	G. Baños	First issue	All pages
1.1 23/01/2019	G. Baños	Section name corrected in cover page	Page 1

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEASA)	Miguel Ángel Calero 01/08/2018	Bernhard Mayer 01/08/2018
Stress Methods (TEASP-TL3)	G. Baños 23/01/2019	Florian Glock 01/08/2018

Table 3. *Approval signatures table for last revision of this chapter*